

Hypothesis Testing with E-values - by R. Ramdas and B. Wang

Reading Group 2026

Chapter 9 - FDR control
using compound e-values
Part 2/3

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In the last episode...

Goal testing K different hypotheses

null hypotheses $\rightarrow$ set of probs $\mathcal{P}_1 \dots \mathcal{P}_K$

for $k \in \mathcal{K} := [1, K]$ the k th null is

$$H_k : P^+ \in \mathcal{P}_k$$

In the last episode...

Def 9.1 (Compound e-variable)

Let $E_1, \dots, E_k$ be $[0, \infty]$ -valued r.v.

We say that $E_1, \dots, E_k$ are compound e-variable for $(\mathcal{P}_1, \dots, \mathcal{P}_k)$ if

$$\sum_{k: IP \in \mathcal{P}_k} E^P[E_k] \leq 1 \quad \text{for all } P \in \mathcal{I}_1$$

A vector of e-variable is a compound e-variable, without any restriction on the dependence structure of the e-variable

In the last episode...

Def 9.6 (multiple testing procedure)

A multiple testing procedure $\mathcal{D}$ is a Borel function of X that produces a subset of $\mathcal{H}$ representing the indices of rejected hypotheses

Def 9.7 (False Discovery Rate)

The FDR under $\mathbb{P}$ is

$$\text{FDR}_{\mathcal{D}} = \mathbb{E}^{\mathbb{P}} \left[\frac{1}{|\mathcal{D}|} \sum_{b: P \in \mathcal{D}_b} \mathbb{1}_{b \in \mathcal{D}} \right]$$

In the last episode...

Def 9.8 (The e-BH procedure)

Suppose that $e_1, \dots, e_K$ are realised compound e-values for the hypotheses $(\mathcal{P}_1, \dots, \mathcal{P}_K)$. For $k \in \mathcal{K}$, let $e_{[k]}$ be the k -th order statistics of $e_1, \dots, e_K$, from the largest to the smallest. The **e-BH procedure** at level $\alpha \in (0, 1)$, rejects all hypotheses with the largest k^* e-values, where

$$k^* := \max \left\{ k \in \mathcal{K}, \frac{k e_{[k]}}{K} > \frac{1}{\alpha} \right\}$$

with the convention $\max(\emptyset) = 0$

In the last episode...

Th 9.11 (FDR control)

The e-BH procedure has a FDR control at most $\alpha \in (0, 1)$ ie

$$\text{FDR}_D \leq \alpha \quad \text{for all } P \in \mathcal{M}_1$$

Th 9.18-19 (Universality)

All FDR controlling procedures can be written as the e-BH procedure applied to some compound e-values

9.3 log-optimal simple separable compound e-values

In this section $X = (X_h, h \in \mathcal{H})$

In principle we may test each H_h using only X_h

Def 9.20 (Simple separable e-variable)

$E_1, \dots, E_K$ are separable if for all h , E_h is X_h measurable, that is $E_h = E_h(X_h)$. They are coll simple separable if $E_h = f(X_h)$ for some function f (the same for all h)

9.3.1 Background: Robbins' compound decision theory

Let P_μ be the proba measure

$$\mu = (\mu_1, \dots, \mu_K) \text{ fixed} \quad X_k \sim N(\mu_k, 1) \quad \forall k \in \mathcal{K}$$

Goal construct an estimator $\hat{\mu}_k$ of μ such
that we minimize

$$\frac{1}{K} \sum_{k \in \mathcal{K}} E^\mu [(\hat{\mu}_k - \mu_k)^2]$$

Bayesian view:

If μ_k has a prior distribution B then the best estimator is the Bayes estimator

$$\hat{\mu}_k^B = E^{P_B} [\mu_k | X] = E^{P_B} [\mu_k | X_k]$$

which is simple separable

P_B : the joint distribution of μ and X

Deterministic view

If μ is deterministic. Given knowledge of μ ,
which function $s_\mu: \mathbb{R} \rightarrow \mathbb{R}$ leads to a
simple separable estimator $\hat{\mu}_k^{s_\mu} = s_\mu(x_k)$ the
minimise our risk?

Deterministic
view

fundamental
theorem of
compound
decisions

Bayesian view
with prior

$$\Pi = \frac{1}{K} \sum_{k \in \mathcal{K}} \delta_{\mu_k}$$

$$\mu' \sim \Pi$$

$$X' | \mu' \sim \mathcal{N}(\mu', \Sigma_K)$$

Given any fixed $s: \mathbb{R} \rightarrow \mathbb{R}$

$$\underbrace{\frac{1}{K} \sum_{k \in \mathcal{K}} \mathbb{E}^{\mu} [s(X_k) - \mu]^2]}_{\mu \text{ treated as fixed}} = \frac{1}{K} \sum_{k \in \mathcal{K}} \mathbb{E}^{\mu} [s(X_k) - \mu_k]^2 = \underbrace{\mathbb{E}^{\Pi} [s(X') - \mu']^2]}_{\mu' \text{ random sample from } \Pi}$$

It follows that the optimal simple separable estimator is the Bayes estimator under Π i.e

$$s(x) = \mathbb{E}^{\Pi} [\mu' | X' = x]$$

The idea

- Construct a feasible non separable estimator

$$\hat{\mu}_h = \hat{\mu}_h(x)$$

that has risk close to the optimal simple separable estimator

- Our oracle benchmark is separable
but the feasible estimator doesn't have to

9.3.2 The log-optimal simple separable compound e-value

• For any $S \in \mathcal{M}_1$, let S_k be the k -th marginal of S i.e. the distribution of X_k if $X \sim S$

• Suppose that for all $k \in \mathbb{K}$ the k -th null

$$\mathcal{P}_k = \{ S : S_k = \mathbb{P}_k \}$$

for some prespecified $\mathbb{P}_k$ with density p_k

is simple hypothesis on the k -th marginal

We also consider simple alternative on the marginal

- Assume that there exists at least one possible true data-generating distribution P in $\bigcap_{k \in \mathcal{K}} \mathcal{P}_k$
- For $k \in \mathcal{K}$, we let Q_k be distributions on the space of data $\mathcal{Y}$ with density q_k representing the alternative

Goal: solve the following optimization problem

()

$$\begin{aligned} & \text{maximize} && \frac{1}{K} \sum_{h \in \mathcal{K}} \mathbb{E}^{Q_h} [\log(s(X_h))] \\ & s: \mathcal{Y} \rightarrow [0, \infty] \end{aligned}$$

subject to $s(X_1) \dots s(X_K)$ are compound
e-variable for $P_1 \dots P_K$

Th 9.2.1 The optimal solution to () is given by the likelihood ratio of mixture over null and alternative)

$$s(x) = \frac{\sum_{j \in \mathcal{K}} q_j(x)}{\sum_{j \in \mathcal{K}} p_j(x)}$$

Th 9.22 Suppose that $E_1 \dots E_K$ are tight, simple separable compound e-value (for the same null define as in the previous section). Then there exist densities $q_1 \dots q_K$ such that E_K may be represented as in the previous theorem on the support of $\frac{1}{K} \sum_{k \in K} P_k$ and it is without loss of generality to take $q_1 = \dots = q_K$

ie any tight simple separable compound e-value must be as in Th 9.21

The objective value of the optimisation () is

$$KL(\bar{Q}, \bar{P})$$

By convexity $KL(\bar{Q}, \bar{P}) \leq \sum_{h \in \mathcal{H}} KL(Q_h, P_h)$

ie that the optimal simple separable compound e-value has a worse objective than using the optimal separable e-value for each individual testing problem

P_h v.s Q_h

given by $E_h^* = \frac{q_h(X_h)}{p_h(X_h)}$

9.4 A minimally adaptive e-BH procedure

1) Choose an e-merging function from chapter 8

$$F: [0, \infty)^k \rightarrow [0, \infty)$$

Reminder

Def An e-merging function (of k e-values) is an increasing Borel function

$$F: [0, \infty)^k \rightarrow [0, \infty)$$

such that for any hypothesis, $F(E_1, \dots, E_k)$ is an e-variable for any $E_1, \dots, E_k$

2) Choose $\alpha \in (0, 1)$

The improved e-BH procedure, denoted by D_α^F is designed as follow:

- 1) Test the global null $\bigcap_{h=1}^K P_h$ via the rejection condition

$$F(e_1, \dots, e_h) \geq \frac{1}{\alpha}$$

which as type I error at most α

- 2) If the global null is rejected, apply the e-BH procedure at level

$$\alpha' = \frac{K\alpha}{K-1}$$

ie 1) if $F(e_1, \dots, e_k) < \frac{1}{\alpha}$ then $D_\alpha^F = \emptyset$

2) if $F(e_1, \dots, e_k) \geq \frac{1}{\alpha}$ then $D_\alpha^F = D_{\alpha'}^{e-BH}$

Prop 9.23 For $\alpha \in (0, 1)$, the improved e-BH procedure D_α^F applied to arbitrary e-values has FDR at most α .

In case $F = M_k$, $D_\alpha^{M_k}$ dominates the e-BH procedure D_α^{e-BH} i.e. $D_\alpha \subseteq D_\alpha^{M_k}$